SLEEPJ, 2026, **49**, 1–13
doi:10.1093/sleep/zsaf358
Advance Access publication 13 November 2025
Original Article

Original Article

Performance of wearable light sensors for measuring photopic and melanopic illuminance under laboratory and free-living conditions

Asuka Ishihara*, Robert J. Brychta, Samuel R. LaMunion, Suzanne McGehee, Magaly Donayre, Olivia Jordan, Kyndall Davis, Gabriel Sanchez, Lindsey Smith, Stephanie T. Chung, Amber B. Courville and Kong Y. Chen*

National Institute of Diabetes and Digestive and Kidney Diseases, National Institutes of Health, Bethesda, MD, United States

*Corresponding authors. Asuka Ishihara, National Institute of Diabetes and Digestive and Kidney Diseases, 9000 Rockville Pike, 10 Center Drive 5N323, Bethesda, MD 20892, United States. Email: asuka.ishihara@nih.gov and Kong Y. Chen, National Institute of Diabetes and Digestive and Kidney Diseases, 9000 Rockville Pike, 10 Center Drive 5N323A, Bethesda, MD 20892, United States. Email: chenkong@niddk.nih.gov.

Abstract

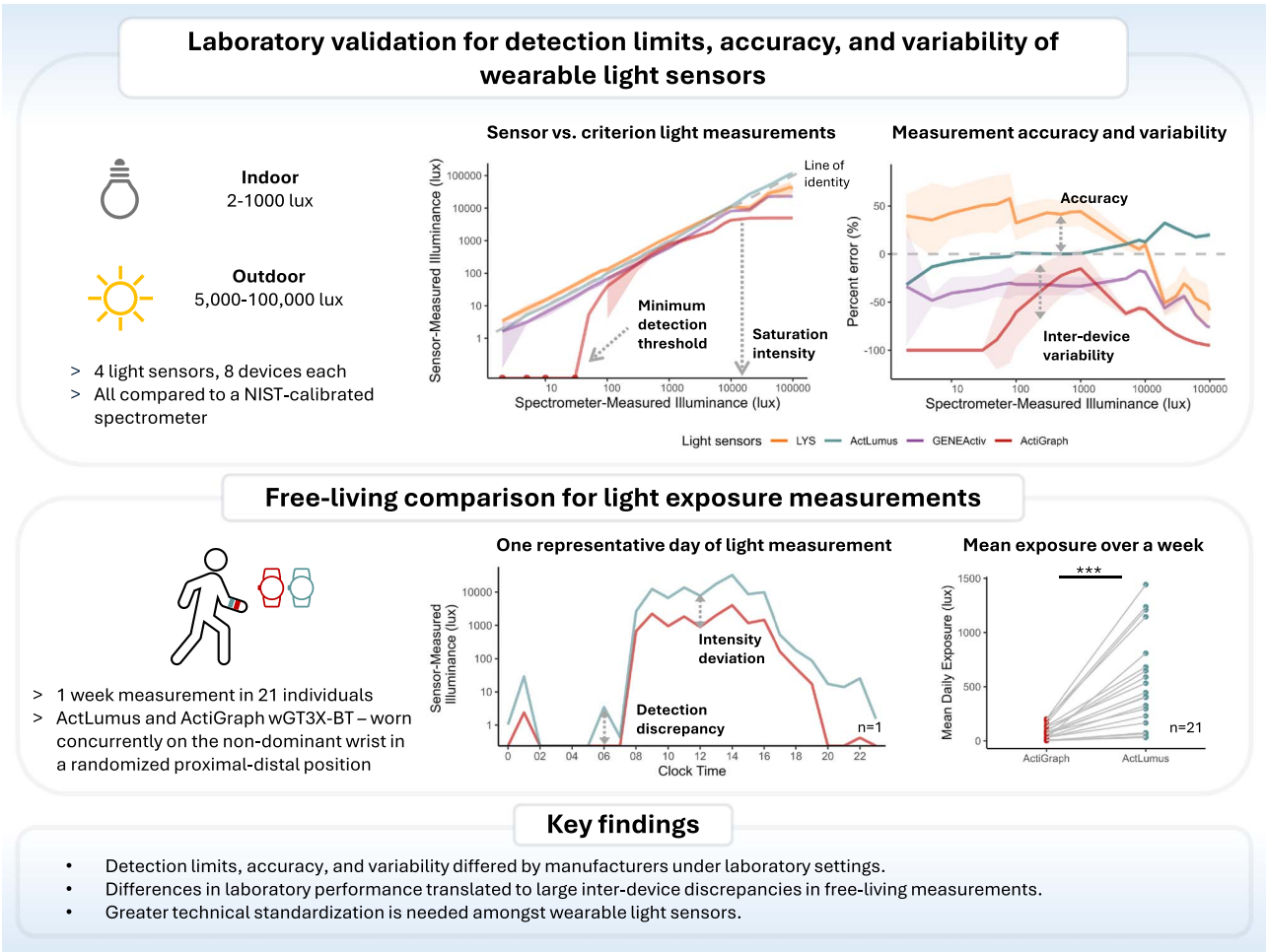
Wearable sensors are commonly used to study the effects of free-living light exposure on physiological outcomes; however, rigorous validation of their performance has been limited. To address this gap, we quantified the accuracy and precision of light measurements from four commonly used wearable sensors (LYS, ActLumus, GENEActiv, and ActiGraph wGT3X-BT). Eight devices from each manufacturer (32 total) were compared to a criterion spectrometer under light-emitting diode–generated cool (peak ∼460 nm) and warm (peak ∼600 nm) white light at indoor intensities (2–1000 lux) and broad-spectrum sunlight (5 k–100 k lux). We found linear responses for all sensors between 50 and 20 k photopic lux, irrespective of spectrum, but detection ranges varied: ActLumus (2–100 k lux), LYS (2–60 k lux), GENEActiv (2–40 k lux), and ActiGraph (50–20 k lux). Accuracy—percent difference to criterion—also differed: ActLumus ($+3.3 \pm 13.4\%$, mean ± SD), LYS ($+19.5 \pm 42.0\%$), and below −20% for GENEActiv and ActiGraph. Interdevice variability was consistently lowest for ActLumus, and melanopic illuminance was less accurate with LYS than ActLumus, particularly in warm indoor light. Laboratory findings were compared to 1-week free-living light measurements in 21 individuals concurrently wearing the most and least accurate sensors in randomized positions on the nondominant wrist. The mean 24 h lux per participant was correlated ($r = 0.91$, $p < .001$) but lower for ActiGraph (77.3 ± 68.5 lux) than ActLumus (515.0 ± 436.0 lux; $p < .001$) with the greatest differences ≤100 lux ($p < .001$), consistent with the laboratory results. Thus, differences in illuminance range and accuracy can lead to large disparities in free-living measures across manufacturers, suggesting a need for greater technical standardization.

Clinical Trial: A Natural History Study of Metabolic Sizing in Health and Disease. https://clinicaltrials.gov/study/NCT05398783 (NCT05398783).

Key words: light exposure; wearables; sensitivity; variability

Submitted: 8 August, 2025; **Revised:** 30 October, 2025; **Accepted:** 3 November, 2025
Published by Oxford University Press on behalf of Sleep Research Society 2025.
This work is written by (a) US Government employee(s) and is in the public domain in the US.

Graphical Abstract



Statement of Significance

Emerging evidence suggests light exposure has a greater impact on human health than previously appreciated. Wearable sensors embedded with light-measuring capabilities can play a crucial role in studying the relationship between light and physiological health. Thus, the performance of light sensors should be thoroughly evaluated. We tested the detection limits, accuracy, and variability of four commonly used wearable light sensors to indoor and outdoor intensities and spectra reflective of the real-world light environment and assessed the translatability of laboratory-based observations in a free-living pilot study. Our findings demonstrate broad variation in accuracy within and across manufacturers that emphasizes the need for rigorous sensor evaluation and highlights the challenges of interpreting illuminance measures across studies deploying different light sensors.

Introduction

Light influences various aspects of human health [1–3], including the entrainment of circadian rhythms and the regulation of sleep-/wake cycles [4]. Laboratory-based studies have demonstrated that the consequences and benefits of light exposure depend on various factors, such as intensity, duration, wavelength, and timing [5–9]. Although these findings provide mechanistic insight into the physiological responses to light in controlled settings, free-living data are also needed to determine how they translate to the daily light environment, where light levels can vary drastically [10]. Self-report can be used to estimate free-living light exposures, but its subjectivity introduces potential biases on exposure durations and intensities [11]. Wearable light sensors, on the other hand, offer a more objective method to quantify real-world light exposure and have been used to study the relationship of photic exposure to physiological health outcomes such as sleep [12], daytime activity [13], type 2 diabetes [14, 15], and cardiometabolic mortality [16].

Despite the advantages sensors offer in collecting objective light information on an individual basis under more naturalistic conditions, few independent studies have systematically investigated the accuracy or precision of these light sensors against a criterion standard measurement [17–21]. Moreover, manufacturer-provided technical information on hardware specifications and performance is typically limited [22]. This lack of comparative detail can complicate interpretation of results obtained with different sensors. For example, one recent meta-analysis found measured light levels of office workers ranged broadly, averaging from

Table 1. Wearable light sensor specifications*

	LYS	ActLumus	GENEActiv	wGT3x-BT
Manufacturer	LYS Technologies	Condor Instruments	Activinsights	ActiGraph
Sensors	Ambient light Accelerometer	Accelerometer Ambient light Skin temperature	Accelerometer Ambient light Temperature	Accelerometer Ambient light
Light measurements	Photopic, Melanopic	Photopic, Melanopic	Photopic	Photopic
Spectral ranges	380–1100 nm (RGB)	~360–1100 nm (10 channels)	Unknown	Unknown
Illuminance ranges (lux)	0–100 000	1–100 000	0–20 000	5000
Accuracy as advised	90%–95% (0–500 lx)	90% (1000 lux)	±90% (1000 lux)	Unknown

*Obtained from the manufacturer manuals and websites.

72 to 1134 lux [23]. However, the nine studies incorporated in the analysis each had distinct designs and included eight different light sensors, making it difficult to determine what proportion of the variation was due to differences in actual light environments versus potential disparities in sensor performance [23].

Although wearable light sensors have traditionally measured photopic illuminance, weighted to match the spectrum of image-forming cone photoreceptors (peak at ~555 nm), some models also estimate melanopic illuminance. This metric reflects the spectral sensitivity of melanopsin (peak at ~480 nm), the photopigment of intrinsically photosensitive retinal ganglion cells that play an important role in the regulation of sleep–wake cycles, circadian processes, and other nonvisual responses governed by the suprachiasmatic nucleus [24–26]. Despite its noted biological relevance [7], to our knowledge, the validity of melanopic output from wearable sensors has not been evaluated.

To better facilitate comparison of light measurements collected using different wearable sensors, greater technical evaluation of their hardware components and software-based algorithms is needed. Thus, we assessed the reliability and validity of four currently available wearable light sensors to standardized stimuli in the lab and under free-living conditions. We designed this investigation as part of the initial verification step proposed in the V3+ framework—a set of guidelines put forth by the Digital Medicine Society to evaluate biometric monitoring technologies for use in clinical research [27, 28]. The primary aim of this study was to quantify the accuracy and variability of these research- and consumer-grade wearable light sensors that measure photopic illuminance in a range of intensities and spectra typical of real-world indoor and outdoor settings. Our secondary aims included evaluating the accuracy and precision of melanopic measurements in two of the devices that provide it and to determine how laboratory findings translate to real-world conditions in a pilot study of simultaneous wear of two devices capturing daily free-living light exposure.

Methods

Laboratory assessment with standardized stimuli

Wearable light sensors. Wearable light sensors from four manufacturers, LYS (LYS Technologies, Copenhagen, Denmark), ActLumus (Condor Instruments, São Paulo, Brazil), GENEActiv (Activinsights, Cambridgeshire, UK), and ActiGraph wGT3X-BT (ActiGraph, Pensacola, FL, USA), were compared against an NIST-calibrated criterion spectrometer (ILT560-A, International Light Technologies, Peabody, MA, USA). These four wearable sensors were selected for their availability at the time of testing and use in ongoing studies in our laboratory and those of our collaborators. Specifications for each sensor are listed in Table 1.

Protocol for technical validation in the laboratory. We conducted studies under a wide range of light intensities and multiple spectra. To test light sensor performance in indoor light (≤80 lux) and outdoor conditions, 32 devices, 8 from each manufacturer, were placed randomly on a flat surface (26 × 42 cm) around a criterion spectrometer located in the center (sample setup shown in Figure 1A). The devices were spaced to mitigate obstruction or reflection from encasings, account for the varied dimensions of each sensor, and ease swapping of devices between trials. For brighter indoor light (100–1000 lux), sixteen sensors (four from each manufacturer) were concurrently tested at a time in each trial to minimize possible deviations from the centrally located spectrometer. When measured with the criterion spectrometer, lux values in the test areas were determined to deviate <10% at all tested intensities, in accordance with 2024 International Electrotechnical Commission testing standards [29] (Figure 1A). All indoor light testing took place in a box with walls covered by light-absorbing, nonreflective black sheets (Edmund Optics, Barrington, NJ, USA) to minimize reflection and maximize uniformity of the projected light. Light-emitting diode (LED bulbs (GE Cync Smart LED, GE Lighting) with adjustable intensity and spectral content were used to generate polychromatic white light with spectra typical of indoor settings, including "cool white" light with a peak at 460 nm and "warm white" light with a peak at 600 nm (Figure 1B). The indoor light trials included six dimmer (2, 5, 10, 30, 50, 80 lux) and five brighter (100, 300, 500, 700, 1000 lux) target intensities of each spectrum. Each indoor intensity was presented in a randomized order for 30 minutes, followed by a 10-minute period of darkness to minimize potential carryover effects (hysteresis). Dimmer indoor light was tested with one light bulb placed approximately 200 cm away from the light sensors. Brighter intensity indoor light trials required two bulbs placed approximately 100 cm away from the light sensors. Full-spectrum outdoor light was recorded from midday to sunset by all sensors placed in an unobstructed location (open field or large courtyard) on both overcast (5 k, 8 k, 10 k lux) and sunny days (20 k, 30 k, 40 k, 50 k, 60 k, 90 k, 100 k lux) between August and November. Sample data from the criterion spectrometer for indoor and outdoor testing conditions are shown in Figure 1C.

Analysis. The LYS and ActLumus devices were programmed to sample every 15 s and 10 s using the LYS Collect application and ActStudio Software (ver. 2.2.0), respectively. The GENEActiv and ActiGraph devices were programmed to collect raw data at 30 Hz, which was later exported in 10-s epoch for analysis using the

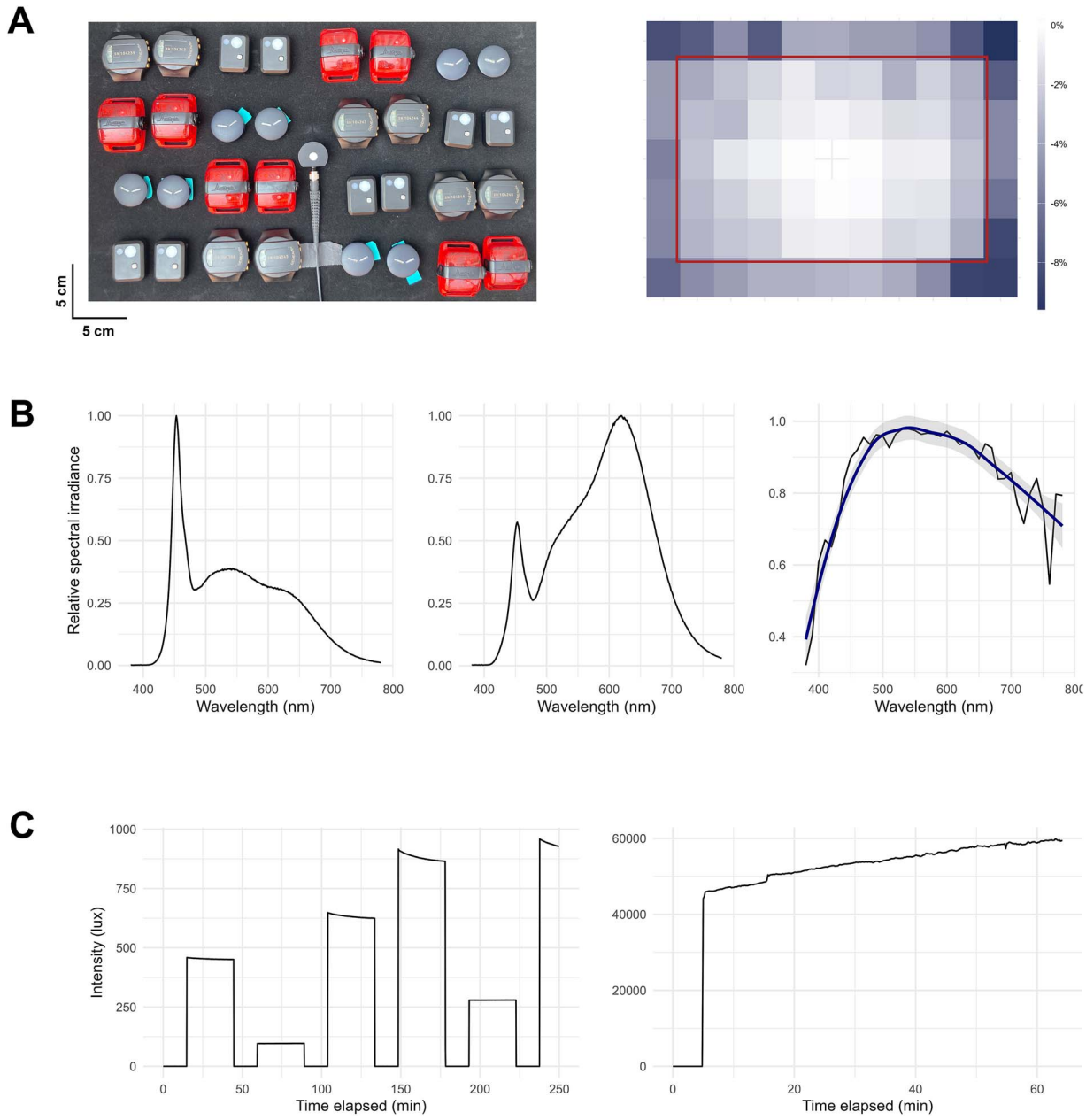


Figure 1. Light sensor testing conditions and sample data. (A) Light sensors were randomly placed surrounding the spectrometer (sample layout left) within the ≤10% deviation range (outlined rectangle) (right). (B) Spectral distribution of the tested light conditions, including cool white (left), warm white (center), and sunlight (right). The sunlight figure shows the measured spectral distribution (solid line) and LOESS smoothing (bold line) with SE (shaded). (C) Sample data from the criterion spectrometer under indoor (left) and outdoor (right) light conditions.

GENEActiv Windows Software from Activinsights (ver. 4.0.9) and ActiLife from ActiGraph wGT3X-BT (ver. 6.13.4), respectively. The mean of a 3-minute steady-state period at each light condition and intensity was used to compare measurements from each device to the criterion, expressed as mean ± SD of all devices per manufacturer. All wearable light sensors provide photopic illuminance values, whereas melanopic illuminance was only available for the LYS and ActLumus sensors. Photopic and melanopic illuminance for the criterion spectrometer was calculated using the conversion factors derived from the International Commission on Illumination (CIE) [30, 31]. The strength of the relationship between the light sensor and spectrometer output was assessed using the marginal R^2 (R^2_m) values from mixed-effects models with device-specific random intercepts, while their agreement was assessed with concordance correlation coefficients (CCCs) with 95% confidence intervals. Modified Bland–Altman plots, with absolute differences to the criterion on the y-axis vs. criterion readings on the x-axis, were used to evaluate the agreement bias between the light sensors and the criterion for each condition (dimmer ≤80 lux, brighter 100–1000 lux, outdoor 5 k–100 k lux). Interdevice variability was determined by the standard deviation of percent error. All data were analyzed and visualized with R studio (ver. 4.2.0).

Free-living comparisons

Twenty-one healthy adult participants (16F/5M; 35.1 ± 15.9 yrs) from an ongoing natural history study of the relationship between body size, composition, and energy metabolism (ClinicalTrials.gov

identifier number: NCT05398783) were recruited to concurrently wear the ActLumus and ActiGraph wGT3X-BT for 1 week in free living. The two wearable sensors were originally selected to measure free-living light exposure, sleep, and physical activity in this protocol. The ActiGraph wGT3X-BT provides raw accelerometer data, employs validated algorithms for step count [32] and sleep scoring [33, 34], and has been incorporated in large-scale, US-based studies [35]. The Condor ActLumus device was newly available to researchers at the time of testing and capable of measuring light intensity in eight separate spectral bands and near-infrared wavelengths (ActLumus manual ver. 2.1.0). These two wearable sensors also had contrasting performance in the laboratory-based test under both indoor and outdoor conditions. Participants were instructed to wear both sensors for 7 days (24 hours/day) on their nondominant wrist, with randomized proximal and distal placement. A rolling accruement of participants ensured a random seasonal distribution of assessment periods, which occurred between mid-April 2024 and May of 2025 (Table S1). Each participant received a unique combination from the 9 available ActLumus and 13 different ActiGraph devices. ActLumus devices were programmed to sample every 60 s using ActStudio (ver 2.2.0). ActiGraph collected raw data at 30 Hz with idle sleep mode disabled, which was later exported in 60-s epochs for analysis using ActiLife (ver 6.13.4). Periods of ActiGraph nonwear were detected using a modified version of Troiano's method [36]—zero activity counts in all three cardinal axes for ≥60 minutes. Output from the manufacturer-described "off-wrist" capacitance sensor was used to detect ActLumus nonwear. Only periods of simultaneous wear on days with ≥10 h [36] of concurrent wear were used for analyses. Pearson's correlation was applied to assess the relationship between the 24-hour mean lux outputs for the two devices in the free living. Paired Wilcoxon signed-rank test was used to compare the percent of wear time spent in four illuminance categories (0, 1–100, 100–1000, >1000 lux).

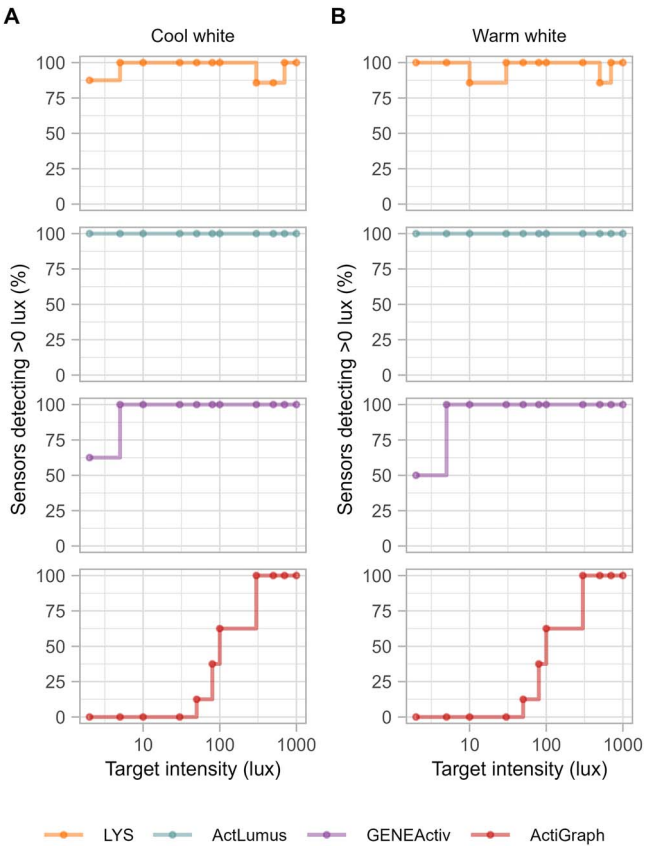


Figure 2. Light sensor detection limits under cool and warm indoor white light. Percentage of devices from each manufacturer detecting lux output >0 lux under target illuminance below 1000 lux in (A) cool and (B) warm indoor light.

Results

Detection limits and linearity for photopic illuminance

We first describe the performance of each wearable light sensor in terms of its minimum detection threshold, linearity, and saturation intensity.

All tested indoor light intensities were detected by 100% of ActLumus devices (Figure 2A and B). Over 85% of LYS devices also detected light at each intensity. About half of the GENEActiv devices did not detect any light at 2 lux, but 100% of the devices detected all other photopic intensities (≥5 lux). No ActiGraph wGT3X-BT devices detected light ≤30 lux, one device registered a non-zero value at 50 lux, and 100% device detection was not achieved until 300 lux. Results for all sensors were largely similar for cool (Figs. 2A and S1A) and warm (Figs. 2B and S1B) white light.

Intermittent dropout—an absence of lux measurements during the 3-minute period sampled for analysis—was observed for all but three LYS devices at various intensities (Figure S1). We excluded one LYS device with consistent dropout from further analysis (LYS2 in Figure S1). None of the other light sensors had any dropouts during the data collection period (Figure S1).

The output of devices from each manufacturer was plotted against the criterion spectrometer measurements to examine linearity and correlation (Figure 3A, Figure S2). ActLumus device measurements were linear and highly correlated to the criterion spectrometer values throughout the tested intensity range (R^2_m > 0.99 for all testing conditions), with no observable saturation (Figure S2). The output of GENEActiv and LYS were both linear with the criterion below a plateau range (spectrometer values of 40 k–60 k lux). Above this range, as source light intensity increased, the mean lux values remained relatively consistent and below the criterion (Figure 3A, Table S2). Thus, for both sensors, correlation to the criterion was weaker in outdoor than in indoor conditions (Figure S2). The ActiGraph wGT3X-BT had a narrower linear response range, between ~50 and ~20 k lux, with most ActiGraph wGT3X-BT devices outputting 0 lux at criterion intensities <100 lux and all wGT3X-BT devices saturating at ~20 k lux (Figure 3A and S2, Table S2). Consequently, ActiGraph wGT3X-BT showed weaker correlations in dimmer indoor (R^2_m = 0.22 for cool and R^2_m = 0.21 for warm white) and outdoor light R^2_m = 0.37) than in brighter indoor light (R^2_m = 0.85) (Figure S2). The range of linearity and correlation for each manufacturer was largely consistent in warm and cool indoor light (Figures 3A, S2, and S3).

Accuracy of photopic illuminance between light sensors and criterion

The agreement of each light sensor with the criterion spectrometer was further evaluated using the percent difference at each tested intensity (Figure 3B), CCCs for dimmer indoor, brighter indoor, and outdoor conditions (Figure 4), and modified Bland–Altman plots (Figure S4). Overall, ActLumus had the smallest average percent error across all intensities (3.3 ± 13.4%), while LYS (19.5 ± 42.0%), GENEActiv (−35.7 ± 24.2%), and ActiGraph wGT3X-BT (−68.0 ± 37.1%) over/under-reported by ~6–20 fold compared

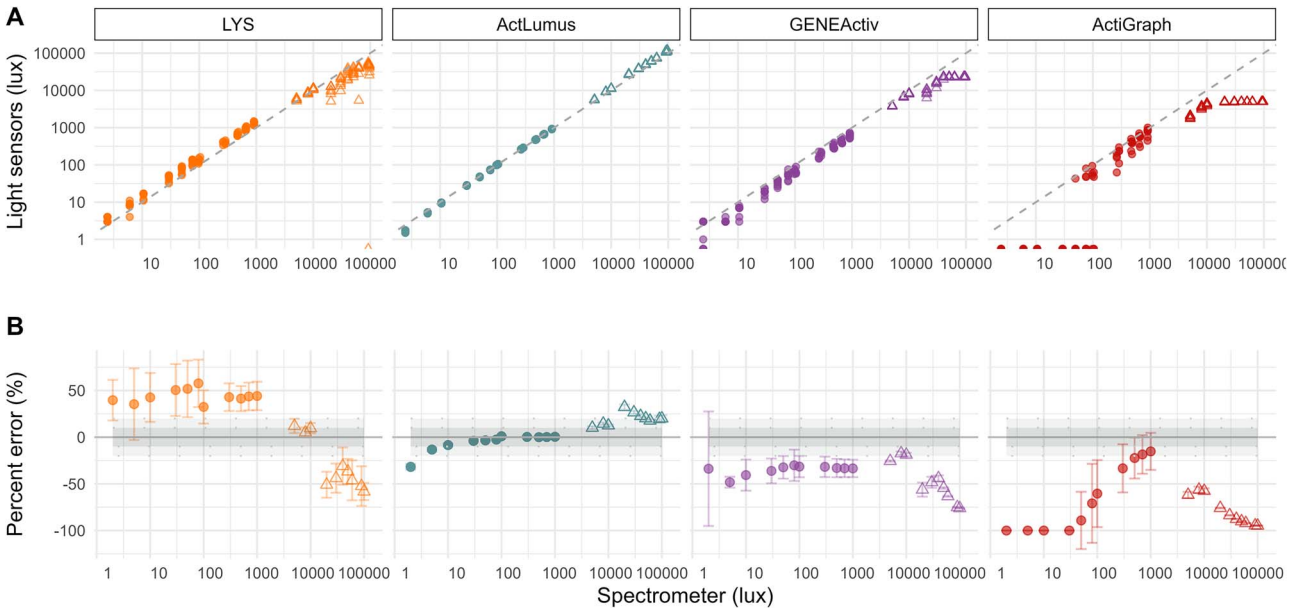


Figure 3. Photopic light measurement accuracy by a wearable sensor. (A) Photopic illuminance for each sensor plotted against the criterion spectrometer. Individual points represent an average over a stable 3-minute period from one device at each target intensity. Symbols represent warm white (circle) and outdoor (triangle) light conditions. Dashed line is the line of identity. (B) Deviation from criterion, shown as mean percent error ± SD of all devices at each target intensity. Darker shade representing ±10% and lighter shade representing ±20% range. Warm white (circle), outdoor light (triangles).

to ActLumus. However, each sensor demonstrated different patterns of accuracy and agreement over the tested intensity range.

While the concordance of LYS and the criterion was high in both dimmer and brighter indoor conditions (CCC > 0.75) (Figure 4), the mean differences were + 46.8 ± 25.4% and + 37.7 ± 15.0% (mean ± SD%), respectively (Figure 3B, Figure S4). In outdoor conditions, LYS under-reported compared to and had lower agreement with the criterion (−30.9 ± 29.1%, CCC = 0.5) (Figure 3B, Figure 4), partially due to an apparent saturation near 60 k lux (Figure S2). The difference to the criterion, however, was smallest (<20%) between 5 k and 10 k lux of outdoor light (Table S2).

ActLumus readings were generally within 20% of the criterion for indoor conditions, with minimal differences in brighter indoor light (0.8 ± 0.9%) (Figure 3B, Table S2). Although ActLumus devices tended to under-report in dimmer indoor light (−8.3 ± 9.0%) and over-report in outdoor light (+19.6 ± 6.6%), high concordance was observed in all three conditions (CCC > 0.95) (Figure 4).

The agreement for GENEActiv was moderately high in dimmer (CCC > 0.8) and brighter (CCC > 0.7) indoor light conditions. For ActiGraph wGT3X-BT, concordance was high in brighter (CCC > 0.85) but poor for dimmer indoor light (CCC > 0.2) due to low detectability below 300 lux (Figures 4 and S2). Means for GENEActiv and ActiGraph wGT3X-BT were consistently below the criterion for both dimmer (−31.2 ± 30.6%; −93.1 ± 23.6%) and brighter indoor conditions (−28.8 ± 10.7%; −26.6 ± 30.4%), respectively (Figures 3B and S4, Table S2). Agreement was low for both ActiGraph wGT3X-BT (CCC = 0.02) and GENEActiv (CCC = 0.24) in outdoor conditions (Figure 4), as both sensors under-reported by a respective −79.4 ± 14.9% and −47.9 ± 20.8% due to their maximum saturation intensities (Figure 3B and S4). Interestingly, although the manufacturer-specified maximum detectable illuminance was 5 k lux for ActiGraph wGT3X-BT and 20 k lux for GENEActiv, they did not reach these maximum values until spectrometer-measured intensities above 20 k and 40 k lux, respectively (Table S2).

Performance for indoor cool and warm white light was comparable for all light sensors across intensities (Figure 3 and S3).

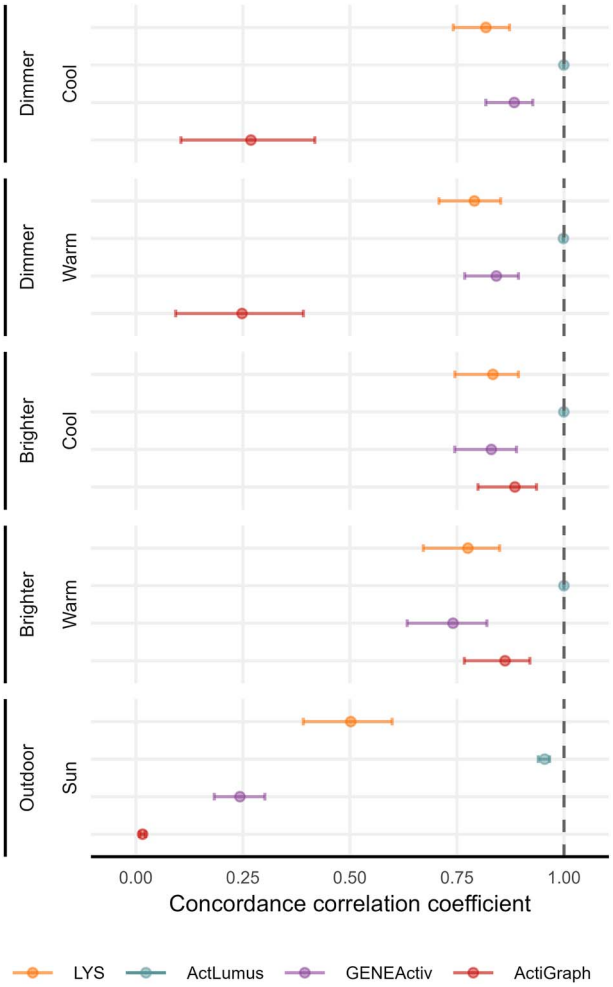


Figure 4. Concordance correlation coefficient (CCC) for photopic illuminance. Concordance with the criterion over all devices per manufacturer shown with 95% confidence intervals for dimmer indoor, brighter indoor, and outdoor conditions.

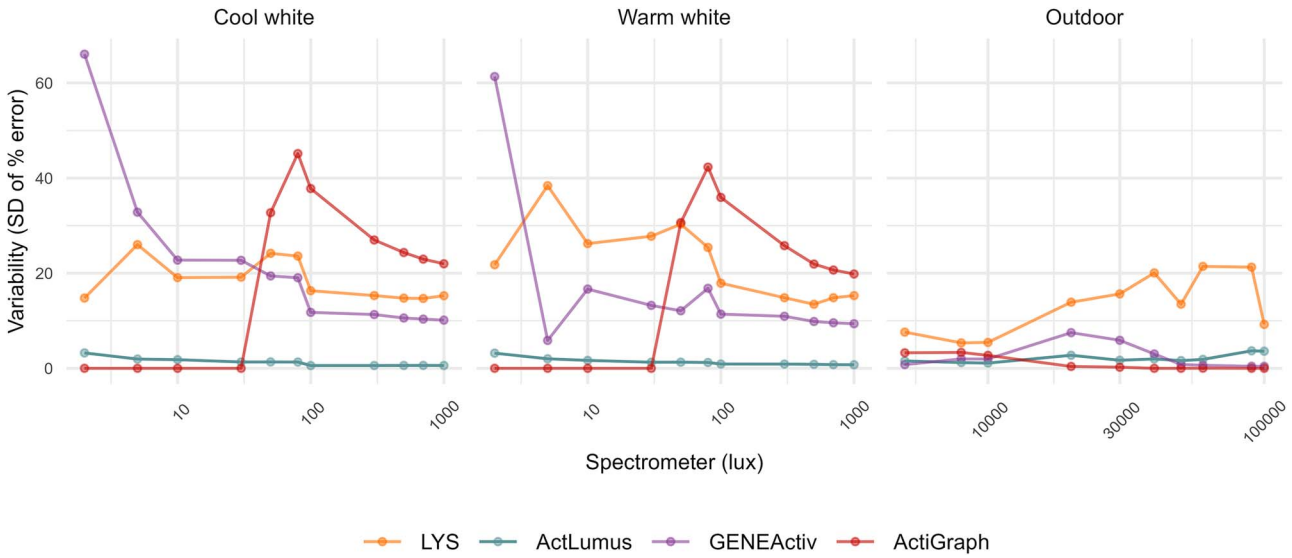


Figure 5. Inter-device variability of each light sensor. Variation shown as the standard deviation of the percent error for photopic illuminance for each light sensor under cool indoor light, warm indoor light, and outdoor light. Variability amongst ActiGraph devices was 0 at low illuminance when all devices registered 0 lux (<50 lux) and high illuminance when all devices reached the detectable threshold (>20 k lux).

Interdevice variability of photopic illuminance

To quantify interdevice variability, we calculated the standard deviation of the percent error of all devices per manufacturer at each tested intensity (Figure 5). The between-device variability was smallest for ActLumus across all intensities: 1.8% for dimmer indoor, 0.7% for brighter indoor, and 2.1% for outdoor conditions. For both LYS and GENEActiv, variability was greater in dimmer (24.7% and 25.7%, respectively) than brighter (15.3% and 10.5%, respectively) indoor light. ActiGraph interdevice variability was <5% at intensities below its detectable threshold (50 lux) but was higher than all other sensors at 25.8% from 80 to 1000 lux. In outdoor light exceeding their respective saturation intensities, Actigraph and GENEActiv exhibited low variance as all device readings plateaued at a manufacturer-specific value. The mean output of LYS devices also appeared to plateau at higher outdoor intensities, but between-device variability increased for LYS at spectrometer-measured values >10 k lux, suggesting the saturation output values for LYS devices may vary more, or output at high intensities was less consistent than those of other sensors (Figure 5). For each manufacturer, the pattern of interdevice variability across indoor intensities was largely similar in cool and warm white light (Figure 5).

Accuracy and variability of melanopic illuminance

We also compared the accuracy of melanopic illuminance, which was available only for LYS and ActLumus sensors. Melanopic illuminance measures from both manufacturers increased linearly with the criterion across all intensities (Figure 6A). ActLumus demonstrated excellent agreement with the criterion (CCC > 0.93, Figure 6C) and deviations for indoor light were generally <20% (Figure 6B), with a small difference between cool (−4.3 ± 7.9%) and warm white light (−5.4 ± 17.9%) (Figure 6C). Outdoor light, however, was over-reported (+23.1 ± 6.5%) (Figure 6B and S5).

Accuracy for LYS melanopic illuminance depended largely on the spectral content of the light source. Deviations were generally within 20% in cool white light (−10.9 ± 23.7%, CCC > 0.85, Figure 6B and C), while warm white light was over-reported (+209.4 ± 111.7%, CCC > 0.2, Figure 6B and C). Concordance for LYS melanopic illuminance was also low in outdoor light (CCC = 0.36, Figure 6C), with illuminance levels between 5 k and 50 k lux being under-reported (−46.5 ± 21.2%, Figure 6B and C) and >50 k lux being over-reported (+73.3 ± 156.7%, Figure 6B and C) (Table S3).

As with photopic illuminance, interdevice variability for melanopic illuminance was considerably higher for LYS than ActLumus (Figure 6D). The standard deviation of the percent error from ActLumus devices was generally below 5% across all intensities, which was consistent in both cool and warm white light and sunlight. For LYS, interdevice variability differed with spectral composition and illuminance. LYS variability in cool white indoor lighting ranged from 18% to 37% for dimmer and 9% to 13% for brighter conditions. The LYS between-device dispersion was higher in warm white indoor lighting, ranging from 80% to160% in dimmer and 46% to 94% in brighter conditions. The smallest variability for LYS devices (<20%) was observed between 5 k and 40 k lux but exceeded 60% in outdoor light above 50 k lux.

Free-living comparisons

To better understand how differences observed in the standardized laboratory conditions are reflected in free-living measurements, we compared the photopic lux outputs from concurrently worn ActLumus and ActiGraph wGT3X-BT (Table S1). As in the laboratory tests, intensity measurement from the ActiGraph wGT3X-BT was drastically lower than that from the ActLumus (Figure 7A) across all individuals—with daily mean ± SD lux values of 77.3 ± 68.5 lux and 515.0 ± 436.0 lux ($p < .001$), respectively. Despite the incongruent absolute lux output, the two sensors showed similar temporal response patterns when ActLumus-measured lux was >100 (Figure 7A and B) and their mean daily lux values per participant were well correlated ($R = 0.91$, $p < .001$) (Figure 7C).

The degree of agreement and correlation between the sensors was dependent on light intensity. The time spent in each intensity bracket during the wear period was significantly different between the two devices (Figure 7D), with ActiGraph showing a greater percentage of wear-time spent at 0 lux than ActLumus (88.5 ± 7.6% vs. 39.7 ± 10.3%, $p < .001$) and ActLumus showing a greater percentage of time spent in all other intensity ranges

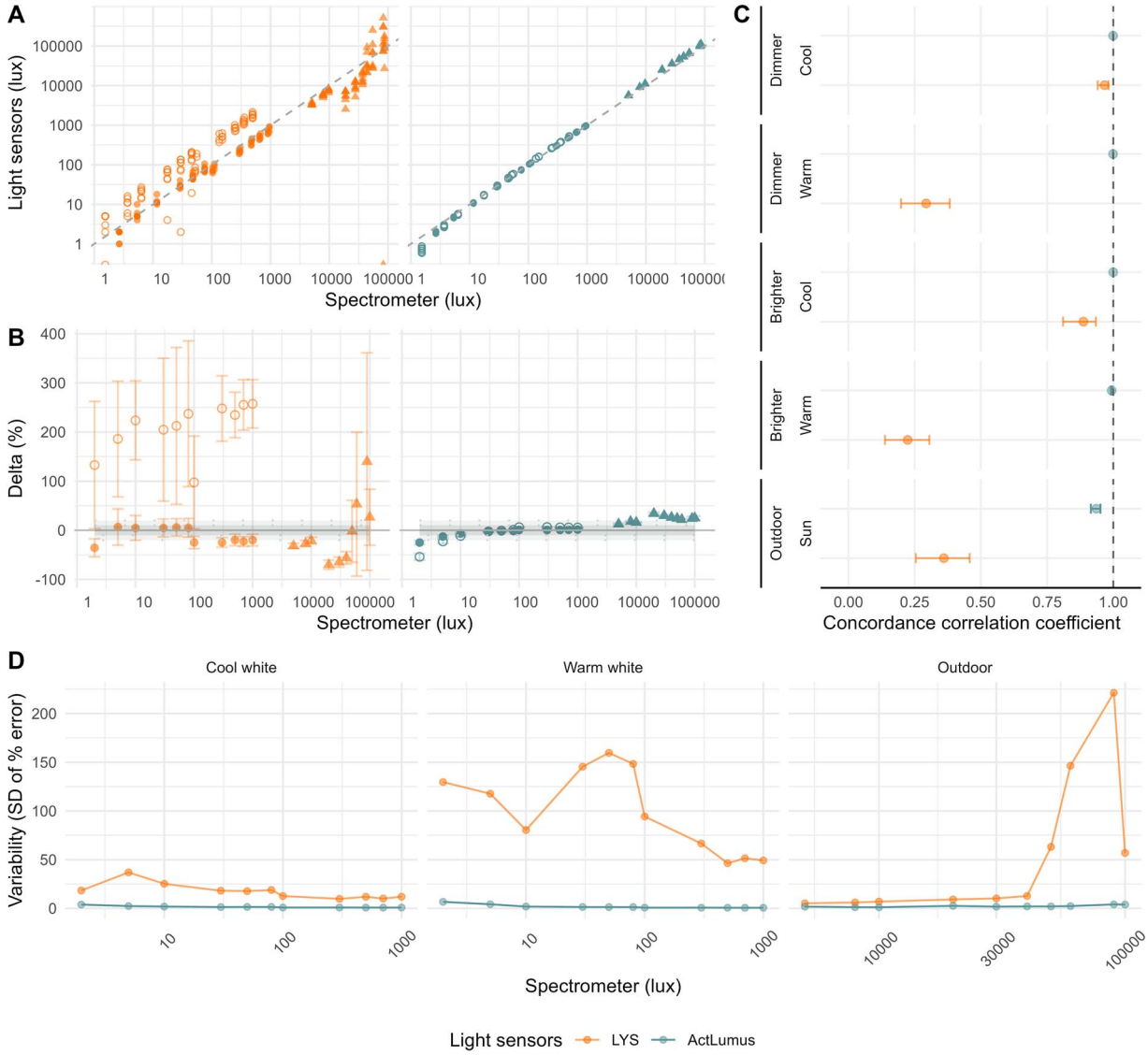


Figure 6. Melanopic light measurement accuracies by wearable sensors. (A) Melanopic illuminance of LYS and ActLumus plotted against the criterion photometer, each symbol represents an average over a stable 3-minute period from one device at each target intensity, dashed lines represent the line of identity. (B) Deviation from the criterion shown as mean percent error ± SD of all devices ($n = 8$) for indoor cool white (closed circle), indoor warm white (open circle), and outdoor light (triangle). The dark-shaded region indicates ±10% and the light-shaded region represents ±20% of the criterion. (C) Concordance correlation coefficient (CCC) with 95% confidence interval. (D) Standard deviation of percent error for cool, warm, and outdoor light conditions.

(Figure 7D). Aside from minutes of 0 lux, the greatest discrepancy between ActiGraph and ActLumus occurred in dimmer indoor light (1–100 lux; 4.6 ± 3.1% vs. 40.8 ± 11.2 %; $p < .001$) (Figure 7D), which agrees with the laboratory finding of a lower minimal detection limit for ActLumus and a lack of sensitivity in dimmer light for ActiGraph devices. Differences among the sensors at higher intensities were less pronounced, though still significant (e.g. >1000 lux, 2.5 ± 2.3% for ActiGraph vs. 4.6 ± 3.6% for ActLumus, $p < .001$) (Figure 7D). Similarly, while minutes spent in higher intensities (>100 lux) registered by the two sensors were correlated, those in lower intensities (1–100 lux) were not (Figure S6A and B).

Discussion

With growing interest in the use of wearable sensors to study the impacts of free-living light exposure on human health, we tested their performance across a range of intensities and spectra typical of real-world light environments. The sensors tested here demonstrated intensity-dependent differences in mean accuracy and interdevice variability of photopic and melanopic illuminance compared to an NIST-calibrated criterion light spectrometer. Detection of light under indoor conditions (≤30 lux) was largely achieved with GENEActiv, LYS, and ActLumus devices but not ActiGraph. Similarly, all sensors that we tested accurately discriminated outdoor from indoor light at intensities above 20 k lux, but performance between 5 k and 10 k lux was mixed and linearity was limited by detector saturation for ActiGraph (~20 k lux), GENEActiv (~40 k lux), and LYS (~60 k lux). The accuracy of melanopic lux varied more with the spectral content of the light source for LYS than for ActLumus. Differences in interdevice variation between manufacturers were broad for both photopic and melanopic illuminance. Simultaneous free-living light measurements confirmed discrepancies in photopic lux output from

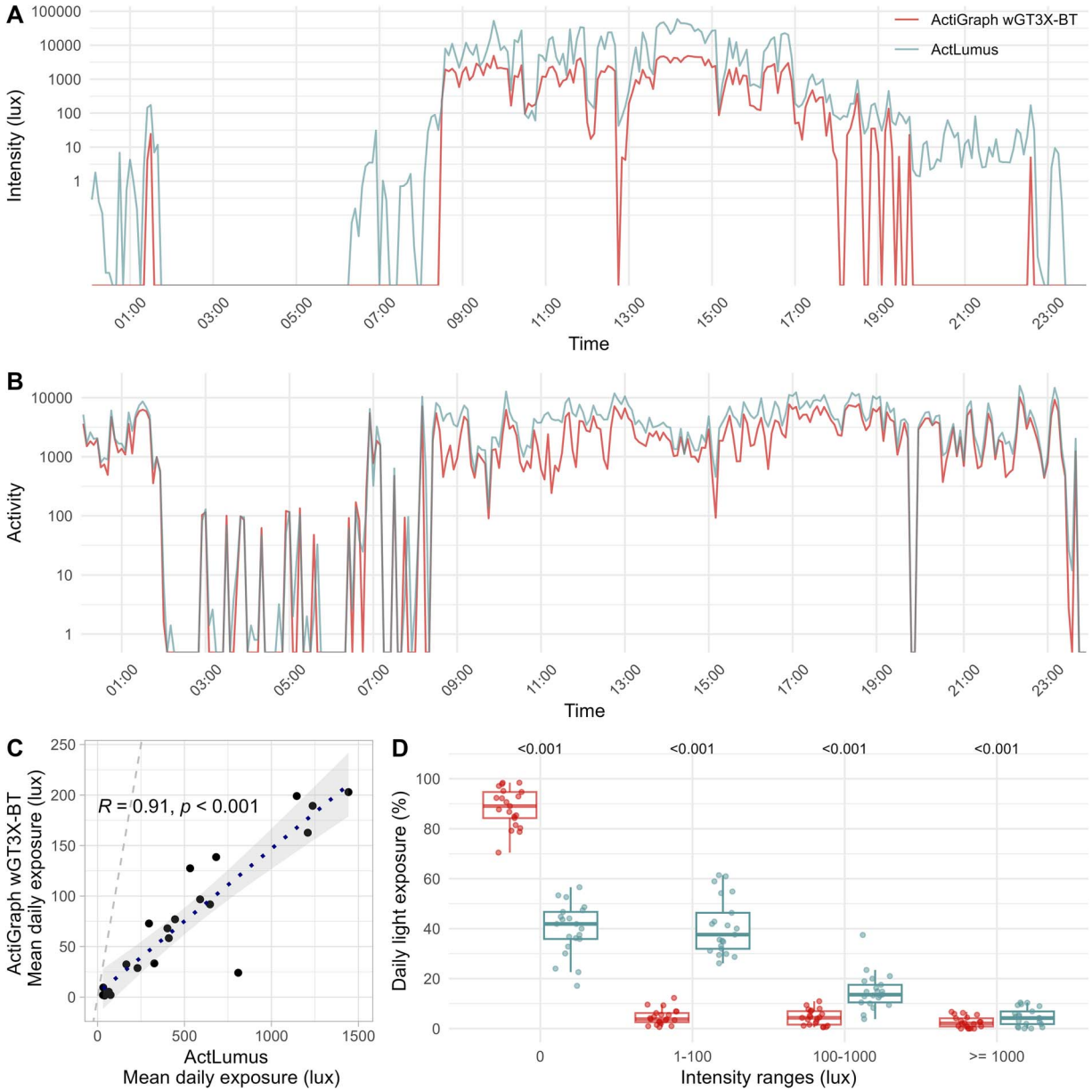


Figure 7. Free-living 24-h cumulative lux measurement between ActLumus and ActiGraph. Sample data from one individual measuring (A) 24-h light exposure in lux and (B) activity in proportional integrating measure (PIM) for ActLumus and vector magnitude for ActiGraph. Low activity levels during the period prolonged of 0 lux (~02:00–~08:00) suggest device wear while sleeping in a dark environment. (C) Pearson correlation (dotted blue line) between 24 h mean light exposure between ActiGraph wGT3X-BT and ActLumus. Individual black dots represent the 24 h mean lux output throughout the entire wear period per individual; the dashed black line is the line of identity. (D) Time spent in each intensity range during the wear period for Actigraph (red) and ActLumus (blue) at each intensity range. Each dot is the mean per individual. Boxes represent the first and third quartiles with the central line indicating the group median. Whiskers are 1.5 times the IQR from the first and third quartiles.

different devices. These findings underscore the importance of considering the illuminance range, photopic and melanopic illuminance accuracies, and interdevice variability required to quantify a specific outcome when selecting a wearable light sensor.

Studies of humans and animals have yielded insight into both the adverse health outcomes and treatment opportunities of light exposure [24, 37]. However, challenges remain in quantifying light measurements accurately in extreme illuminance levels (e.g. very dim light before bedtime or very bright daylight). Studies have demonstrated that nocturnal dim light (<10 lux) alters sleep architecture [38] and exposure to low levels of light (<30 lux) prior to bedtime results in a ~50% suppression of melatonin (albeit with high individual variability) [39], suggesting that detection of this intensity range by wearable sensors may be important for the study of the effects of light on free-living sleep. We found that all devices except the Actigraph sensed light >2 lux. The ActiGraph wGT3X-BT devices detected less than half of the nonzero lux values at intensities between 50 and 100 lux and did not detect any light below 50 lux, limiting its utility for quantifying the presence or absence of light in many built environments. Our free-living data also support the laboratory observations—ActiGraph sensor reported ~50% more minutes of data measured at 0 lux and ~40% fewer minutes between 1 and 100 lux than ActLumus. These findings suggest that differences in minimal detection limit and accuracy under low-intensity light might result in divergent interpretations for outcomes involving dim and moderate light exposure.

While evening light exposure is linked to negative health outcomes, exposure to high intensities of naturalistic daylight is associated with improved mood and sleep [11], circadian rhythm [40], and cognitive performance [41]. The potential benefits of brighter light exposure during the day have been studied by using a light intensity threshold as a proxy to categorize indoor vs. outdoor light [42, 43]. In our laboratory study, we found that under 20 k lux, GENEActiv and ActiGraph wGT3X-BT devices consistently underreported, while LYS devices routinely overreported photopic illuminance compared to the criterion. This divergent accuracy among sensors suggests a common threshold would lead to different time spent in outdoor versus indoor conditions. The results of our free-living study support this observation, as Actlumus and ActiGraph sensors recorded significantly different time above the commonly used 1000 lux threshold [43]. However, despite this difference, time above 1000 lux for ActiGraph and ActLumus was strongly correlated across participants, suggesting each sensor is capable of differentiating outdoor from indoor illuminance, but may require individualized thresholds. Thus, further study is needed to determine whether sensor-specific thresholds may yield more uniform categorization of outdoor exposure time across manufacturers.

For outcome metrics involving cumulative light exposures, properties such as accuracy and range of linearity can affect the comparability of the data between manufacturers. The accuracy of photopic illuminance was found to vary dramatically by sensor and illuminance level. Overall, ActLumus had the lowest error compared to other manufacturers across all intensities. The light intensity range yielding the highest accuracy also differed by sensor—dimmer and brighter indoor light (10–1000 lx) for ActLumus, brighter indoor light (300–1000 lx) for ActiGraph wGT3X-BT, and outdoor light (5 k–10 k lux) for LYS and GENEActiv. Accuracy was also related to the range of linearity for each manufacturer since error increased as source light intensity deviated further from the lower and upper detection limits. The ActiGraph wGT3X-BT devices had the smallest range of linearity (~50–20 k lux), whereas GENEActiv and LYS sensors demonstrated saturation effects at higher intensities (~40 k and ~ 60 k lux, respectively) and ActLumus showed minimal saturation—consistent with a prior model [18]. These differences in accuracy and linearity across photopic intensities were also reflected in our free-living portion of the study comparing cumulative illuminance between ActLumus and ActiGraph. Although 24-h mean lux output from the two devices was highly correlated, ActiGraph output was consistently an order of magnitude smaller than that from ActLumus. The discrepancies in absolute daily mean lux values between sensors may lead to inconsistent recommendations of daily light exposure.

While differences in accuracy illustrate potential issues for comparing data across manufacturers, metrics such as interdevice variability and recording consistency can be useful to gauge confidence when comparing measurements from a batch of devices of the same manufacturer. We found that the standard deviation of percent error varied widely by manufacturer, particularly in indoor light ≤100 lux, light levels in which the majority of the waking day is spent [44], as confirmed by our free-living data. We mitigated the potential contribution of device placement on interdevice variability in our laboratory experiment by randomizing device placement between conditions and limiting the test area to the region where spectrometer-measured light intensity varied by <10%. Reliable, uninterrupted measurement is also crucial to understand how free-living exposures may influence sleep and circadian rhythms. We found that ActLumus, GENEActiv, and ActiGraph wGT3X-BT all provided valid continuous recordings during 1–6 hours of testing. LYS devices were reliable 90% of the time, with occasional dropouts, which were more frequent for some devices than others. While it is difficult to definitively identify the cause of the dropout in our study, we should note that LYS devices have limited on-board memory and rely on Bluetooth communication to regularly transmit data to the software application whereas other sensors do not. Regardless of the cause, the frequency of LYS dropout suggests additional attention is required to confirm the continuity of data collection. Together, these results indicate that most devices can provide continuous free-living recordings, but dropout and interdevice variability may add to the variation between subjects, depending on the sensor and/or measured light conditions.

LYS and ActLumus light sensors both provide measures of melanopic light intensity, which is thought to be more biologically relevant than photopic illuminance since it better predicts the effects of light on the melanopsin-expressing, intrinsically photosensitive retinal ganglion cells responsible for entraining the circadian rhythm [26]. Melanopsin has a peak sensitivity to the shorter wavelengths of light that is reflected in the spectrum used to estimate melanopic illuminance [26]. While the accuracy of melanopic lux from both devices was comparable for dimmer cool white light, ActLumus was much more consistent for the warm spectrum, across intensities, and between devices. The higher melanopic values for LYS in warm white—which includes greater spectral power in the longer wavelengths—suggest an overly broad melanopic filter bandwidth and that the number of spectral channels used to derive melanopic lux (3 RGB channels for LYS versus 10 channels for ActLumus) may play an important role in its accuracy.

Differences in hardware components and calibration procedures likely contribute to the variation in accuracy and variability between manufacturers. According to manufacturer-provided information, most sensors are calibrated with brighter indoor light (500–1000 lux), suggesting higher accuracy compared to the extreme intensities. The noted calibration processes also differed by manufacturer, increasing the potential for inter- and intramanufacturer variation [22]. Differences in photodetectors [22, 45], use of optical filters [18], diffusers [45], and, in some instances, sensor casing [18] likely affected accuracy and interdevice variability. Data quality may also have been influenced by the manufacturer-specific methods of data acquisition/transmission, as mentioned previously. The variability between manufacturers emphasizes the importance of reporting device specifications, both by manufacturers and in future research involving light measurements [46].

In our laboratory testing, ActLumus devices had the broadest detection range, the highest accuracy compared to the criterion spectrometer, and the smallest interdevice variability for both photopic and melanopic illuminance. ActLumus devices also registered the extreme low and high intensities in our free-living measurements, suggesting the expanded detection range may be important to metrics of cumulative light exposure. However, though ActLumus has the capacity to collect other physiological data, it is a relatively new device with fewer validations and published free-living studies than some of the other tested sensors. Contrarily, while ActiGraph wGT3X-BT and GENEActiv have been extensively validated and implemented in large-scale studies of physical activity and sleep [32, 47, 48], their light sensors demonstrated limited performance. Although the ActiGraph wGT3X-BT light sensor exhibited the narrowest response range,

a similar ActiGraph device was observed to differentiate indoor from outdoor light exposure [42], which may be used to supplement activity and sleep measurements. The GENEActiv devices consistently underreported photopic intensities, but coupling the light information with that from its movement [47, 48] and/or temperature [49], sensors may improve the accuracy of sleep timing, nonwear time, and sedentary behavior metrics. The LYS devices provide melanopic output and can be implemented in free-living studies with flexible attachment sites [21, 50]. However, inconsistencies in data collection between devices and intermittent dropouts need to be carefully monitored. Thus, the sensor performance exhibited in this study should be considered alongside other device capabilities and outcomes of interest.

Our study had some limitations. An LED light source was used for our indoor light conditions. Prior assessments of sensor performance using other light sources, such as fluorescent [20] or monochromatic light [18, 20], have produced slightly different results. For instance, an evaluation of the GENEActiv light sensor demonstrated a similar magnitude of under-reporting of LED intensities to that in our study, but an even greater under-reporting of fluorescent light intensities [20]. When the same sensor was tested with monochromatic red, green, and blue light, there was a noted spectral bias toward greater underestimation of shorter wavelengths [18, 20], which was not apparent in our study, as GENEActiv under-reported polychromatic cool and warm white light similarly. These differences, along with the variety of artificial light sources typically used in everyday indoor environments [51] and for sensor calibration [22] highlight the need for further testing of wearable light sensors with varied source lighting. We were only able to test the four models of wearable light sensors available to our laboratory, and thus, this is not a comprehensive study of the many currently available models [22]. However, the data and analyses provide some guidance for wearable sensor selection and raise awareness of potential light sensor output variation within and between manufacturers. The methodologies described here can also be extended to evaluate other sensors. Although our laboratory test incorporated a wide range of intensities and wavelengths of light experienced in daily life, we did not account for angular responses or concurrent exposures to multiple light sources. Lastly, the spacing between devices may have introduced some error during our laboratory testing, although we attempted to minimize the impact by randomizing their placement and limiting testing to the zone of <10% deviation in the criterion sensor output. Selection of a wearable sensor for free-living measurement also requires other considerations, including storage capacity, data complexity and accessibility, battery life, waterproof rating, cost, capability for other on-board measurements (e.g. motion, temperature, heart rate, event marking), and attachment site (e.g. wrist versus closer to the eye) [22]. Therefore, when selecting wearables for light research, the results of our study on light sensor performance should be weighed alongside the other capabilities required to quantify the desired outcome.

Conclusion

We found that most tested wearable light sensors under known stimuli had a linear response to a range of intensities indicative of indoor light to midday outdoor illuminance (50–20 k). However, sensors differed in their detectable intensity range, photopic and melanopic accuracies, and interdevice variability, especially under extreme low and high intensities. Consistent with the laboratory findings, our free-living study also demonstrated that light exposure outputs varied significantly by light sensors. Thus, along with other device-specific features, wearable light sensor selection should depend on the sensitivity and illuminance range required by the free-living outcomes of interest. As wearable light-sensing technology rapidly evolves, simultaneous assessments of sensor performance is crucial to clearly interpret findings and understand the true impact of different light exposures on human health.

Acknowledgments

The authors would like to thank Drs. Charles E. Matthews, Sarah Keadle, and Pedro F. Saint-Maurice for generously providing LYS sensors, and Drs. Kathleen Merikangas and Gena Glickman for providing critical feedback on our draft. We also thank the clinical center staff for their contributions to patient care. Finally, we thank the participants for their contribution to the study. This research was supported by the Intramural Research Program of the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) within the National Institutes of Health (NIH). The contributions of the NIH author(s) were made as part of their official duties as NIH federal employees, are in compliance with agency policy requirements, and are considered Works of the United States Government. However, the findings and conclusions presented in this paper are those of the author(s) and do not necessarily reflect the views of the NIH or the U.S. Department of Health and Human Services.

Author contributions

A.I., R.J.B., S.R.L., and K.Y.C. designed the laboratory study. A.I. performed data collection and analysis of the laboratory protocol. A.I., R.J.B., S.R.L., and K.Y.C. interpreted the data. S.T.C., A.B.C., and K.Y.C. conceptualized and designed the free-living study. A.I., S.M., M.D., O.J., K.D., G.S., L.S., and A.B.C. collected data for the free-living study. A.I. and R.J.B. processed and analyzed free-living data. All authors contributed to reviewing and editing the final version of the paper.

Supplementary material

Supplementary material is available at *SLEEP* online.

Disclosure statement

Financial disclosure: None declared.

Non-financial disclosure: None declared.

Data availability

The data and source code have been deposited in Open Science Framework (osf.io/97dct).

References

1. Bedrosian TA, Nelson RJ. Timing of light exposure affects mood and brain circuits. *Transl Psychiatry*. 2017;**7**(1):e1017. https://doi.org/10.1038/tp.2016.262
2. Paul S, Brown T. Direct effects of the light environment on daily neuroendocrine control. *J Endocrinol*. 2019;**243**(1):R1–R18. https://doi.org/10.1530/JOE-19-0302
3. Fleury G, Masis-Vargas A, Kalsbeek A. Metabolic implications of exposure to light at night: lessons from animal and human

studies. *Obesity*. 2020;**28**(S1):S18–S28. https://doi.org/10.1002/oby.22807
4. Duffy JF, Czeisler CA. Effect of light on human circadian physiology. *Sleep Med Clin*. 2009;**4**(2):165–177. https://doi.org/10.1016/j.jsmc.2009.01.004
5. Prayag AS, Munch M, Aeschbach D, Chellappa SL, Gronfier C. Light modulation of human clocks, wake, and sleep. *Clocks Sleep*. 2019;**1**(1):193–208. https://doi.org/10.3390/clockssleep1010017
6. Cho Y, Ryu SH, Lee BR, Kim KH, Lee E, Choi J. Effects of artificial light at night on human health: a literature review of observational and experimental studies applied to exposure assessment. *Chronobiol Int*. 2015;**32**(9):1294–1310. https://doi.org/10.3109/07420528.2015.1073158
7. Vetter C, Pattison PM, Houser K, *et al*. A review of human physiological responses to light: implications for the development of integrative lighting solutions. *Leukos*. 2022;**18**(3):387–414. https://doi.org/10.1080/15502724.2021.1872383
8. Zee PC, Goldstein CA. Treatment of shift work disorder and jet lag. *Curr Treat Options Neurol*. 2010;**12**(5):396–411. https://doi.org/10.1007/s11940-010-0090-9
9. Maruani J, Geoffroy PA. Bright light as a personalized precision treatment of mood disorders. *Front Psych*. 2019;**10**:85. https://doi.org/10.3389/fpsyt.2019.00085
10. Bhandary SK, Dhakal R, Sanghavi V, Verkicharla PK. Ambient light level varies with different locations and environmental conditions: potential to impact myopia. *PLoS One*. 2021;**16**(7):e0254027. https://doi.org/10.1371/journal.pone.0254027
11. Burns AC, Saxena R, Vetter C, Phillips AJK, Lane JM, Cain SW. Time spent in outdoor light is associated with mood, sleep, and circadian rhythm-related outcomes: a cross-sectional and longitudinal study in over 400,000 UK Biobank participants. *J Affect Disord*. 2021;**295**:347–352. https://doi.org/10.1016/j.jad.2021.08.056
12. Lok R, Ancoli-Israel S, Ensrud K, Redline S, Stone KL, Zeitzer JM. Timing of outdoor light exposure is associated with sleep-wake consolidation in community-dwelling older men. *Front. Sleep*. 2023;**2**:2. https://doi.org/10.3389/frsle.2023.1268379
13. Shochat T, Martin J, Marler M, Ancoli-Israel S. Illumination levels in nursing home patients: effects on sleep and activity rhythms. *J Sleep Res*. 2000;**9**(4):373–379. https://doi.org/10.1046/j.1365-2869.2000.00221.x
14. Obayashi K, Yamagami Y, Kurumatani N, Saeki K. Bedroom lighting environment and incident diabetes mellitus: a longitudinal study of the HEIJO-KYO cohort. *Sleep Med*. 2020;**65**:1–3. https://doi.org/10.1016/j.sleep.2019.07.006
15. Windred DP, Burns AC, Rutter MK, *et al*. Personal light exposure patterns and incidence of type 2 diabetes: analysis of 13 million hours of light sensor data and 670,000 person-years of prospective observation. *Lancet Reg Health Eur*. 2024;**42**:100943. https://doi.org/10.1016/j.lanepe.2024.100943
16. Windred DP, Burns AC, Lane JM, *et al*. Brighter nights and darker days predict higher mortality risk: a prospective analysis of personal light exposure in >88,000 individuals. *Proc Natl Acad Sci U S A*. 2024;**121**(43):e2405924121. https://doi.org/10.1073/pnas.2405924121
17. Markvart J, Hansen ÅM, Christoffersen J. Comparison and correction of the light sensor output from 48 wearable light exposure devices by using a side-by-side field calibration method. *Leukos*. 2015;**11**(3):155–171. https://doi.org/10.1080/15502724.2015.1020948
18. Price LLA, Lyachev A, Khazova M. Optical performance characterization of light-logging actigraphy dosimeters. *J Opt Soc Am A*. 2017;**34**(4):545–557. https://doi.org/10.1364/JOSAA.34.000545
19. Joyce DS, Zele AJ, Feigl B, Adhikari P. The accuracy of artificial and natural light measurements by actigraphs. *J Sleep Res*. 2020;**29**(5):e12963. https://doi.org/10.1111/jsr.12963
20. Stone JE, McGlashan EM, Facer-Childs ER, Cain SW, Phillips AJK. Accuracy of the GENEActiv device for measuring light exposure in sleep and circadian research. *Clocks Sleep*. 2020;**2**(2):143–152. https://doi.org/10.3390/clockssleep2020012
21. Stevenson S, Suppiah H, Driller M. Research note: the validity and reliability of a novel wearable sensor in measuring light exposure in an indoor and outdoor environment. *Light Res Technol*. 2025;0(0). https://doi.org/10.1177/14771535251338582
22. van Duijnhoven J, Hartmeyer SL, Didikoglu A, *et al*. Measuring light exposure in daily life: a review of wearable light loggers. *Build Environ*. 2025;**274**:112771. https://doi.org/10.1016/j.buildenv.2025.112771
23. de Vries SW, Gkaintatzi-Masouti M, van Duijnhoven J, Mardaljevic J, Aarts MPJ. Recommendations for light-dosimetry field studies based on a meta-analysis of personal light levels of office workers. *Light Res Technol*. 2024;**57**(1):47–70. https://doi.org/10.1177/14771535241248540
24. Brown TM, Brainard GC, Cajochen C, *et al*. Recommendations for daytime, evening, and nighttime indoor light exposure to best support physiology, sleep, and wakefulness in healthy adults. *PLoS Biol*. 2022;**20**(3):e3001571. https://doi.org/10.1371/journal.pbio.3001571
25. Lucas RJ, Peirson SN, Berson DM, *et al*. Measuring and using light in the melanopsin age. *Trends Neurosci*. 2014;**37**(1):1–9. https://doi.org/10.1016/j.tins.2013.10.004
26. Brown TM. Melanopic illuminance defines the magnitude of human circadian light responses under a wide range of conditions. *J Pineal Res*. 2020;**69**(1):e12655. https://doi.org/10.1111/jpi.12655
27. Goldsack JC, Coravos A, Bakker JP, *et al*. Verification, analytical validation, and clinical validation (V3): the foundation of determining fit-for-purpose for biometric monitoring technologies (BioMeTs). *NPJ Digit Med*. 2020;**3**(1):55. https://doi.org/10.1038/s41746-020-0260-4
28. Spitschan M, Smolders K, Vandendriessche B, *et al*. Verification, analytical validation and clinical validation (V3) of wearable dosimeters and light loggers. *Digit. Health*. 2022;**8**:205520762211448. https://doi.org/10.1177/20552076221144858
29. Wearable electronic devices and technologies. *Part 402-3: Performance measurement of fitness wearables - Test methods for the determination of the accuracy of heart rate*. Geneva, Switzerland: International Electrotechnical Commission; 2024.
30. Commission Internationale de l'Eclairage 2019, CIE spectral luminous efficiency for photopic vision. Vienna, AT: International Commission on Illumination (CIE). https://cie.co.at/datatable/cie-spectral-luminous-efficiency-photopic-vision
31. CIE system for metrology of optical radiation for ipRGC-influenced responses to light. CIE S 026:2018. https://cie.co.at/publications/cie-system-metrology-optical-radiation-iprgc-influenced-responses-light-0. 2025
32. Ngueleu AM, Barthod C, Best KL, Routhier F, Otis M, Batcho CS. Criterion validity of ActiGraph monitoring devices for step counting and distance measurement in adults and older adults: a systematic review. *J Neuroeng Rehabil*. 2022;**19**(1):112. https://doi.org/10.1186/s12984-022-01085-5
33. Cole RJ, Kripke DF, Gruen W, Mullaney DJ, Gillin JC. Automatic sleep/wake identification from wrist activity. *Sleep*. 1992;**15**(5):461–469. https://doi.org/10.1093/sleep/15.5.461
34. Sadeh A, Sharkey KM, Carskadon MA. Activity-based sleep-wake identification: an empirical test of methodological issues. *Sleep*. 1994;**17**(3):201–207. https://doi.org/10.1093/sleep/17.3.201

35. Troiano RP, McClain JJ, Brychta RJ, Chen KY. Evolution of accelerometer methods for physical activity research. *Brit J Sport Med*. 2014;**48**(13):1019–1023. https://doi.org/10.1136/bjsports-2014-093546
36. Troiano RP, Berrigan D, Dodd KW, Masse LC, Tilert T, McDowell M. Physical activity in the United States measured by accelerometer. *Med Sci Sports Exerc*. 2008;**40**(1):181–188. https://doi.org/10.1249/mss.0b013e31815a51b3
37. Walbeek TJ, Harrison EM, Gorman MR, Glickman GL. Naturalistic intensities of light at night: a review of the potent effects of very dim light on circadian responses and considerations for translational research. *Front Neurol*. 2021;**12**:625334. https://doi.org/10.3389/fneur.2021.625334
38. Cho CH, Yoon HK, Kang SG, Kim L, Lee EI, Lee HJ. Impact of exposure to dim light at night on sleep in female and comparison with male subjects. *Psychiatry Investig*. 2018;**15**(5):520–530. https://doi.org/10.30773/pi.2018.03.17
39. Phillips AJK, Vidafar P, Burns AC, *et al*. High sensitivity and interindividual variability in the response of the human circadian system to evening light. *Proc Natl Acad Sci U S A*. 2019;**116**(24):12019–12024. https://doi.org/10.1073/pnas.1901824116
40. Wright KP, McHill AW, Birks BR, Griffin BR, Rusterholz T, Chinoy ED. Entrainment of the human circadian clock to the natural light-dark cycle. *Curr Biol*. 2013;**23**(16):1554–1558. https://doi.org/10.1016/j.cub.2013.06.039
41. Boubekri M, Lee J, MacNaughton P, *et al*. The impact of optimized daylight and views on the sleep duration and cognitive performance of office workers. *Int J Environ Res Public Health*. 2020;**17**(9):3219. https://doi.org/10.3390/ijerph17093219
42. Flynn JI, Coe DP, Larsen CA, Rider BC, Conger SA, Bassett DR. Detecting indoor and outdoor environments using the ActiGraph GT3X+light sensor in children. *Med Sci Sports Exerc*. 2014;**46**(1):201–206. https://doi.org/10.1249/MSS.0b013e3182a388c0
43. Dautovich ND, Schreiber DR, Imel JL, *et al*. A systematic review of the amount and timing of light in association with objective and subjective sleep outcomes in community-dwelling adults. *Sleep Health*. 2019;**5**(1):31–48. https://doi.org/10.1016/j.sleh.2018.09.006
44. Scheuermaier K, Laffan AM, Duffy JF. Light exposure patterns in healthy older and young adults. *J Biol Rhythm*. 2010;**25**(2):113–122. https://doi.org/10.1177/0748730410361916
45. Danilenko KV, Stefani O, Voronin KA, *et al*. Wearable light-and-motion dataloggers for sleep/wake research: a review. *Appl Sci*. 2022;**12**(22):11794. https://doi.org/10.3390/app122211794
46. Spitschan M, Kervezee L, Lok R, McGlashan E, Najjar RP, ENLIGHT Consortium. ENLIGHT: a consensus checklist for reporting laboratory-based studies on the non-visual effects of light in humans. *EBioMedicine*. 2023;**98**:104889. https://doi.org/10.1016/j.ebiom.2023.104889
47. Antczak D, Lonsdale C, Del Pozo CB, Parker P, Sanders T. Reliability of GENEActiv accelerometers to estimate sleep, physical activity, and sedentary time in children. *Int J Behav Nutr Phys Act*. 2021;**18**(1):73. https://doi.org/10.1186/s12966-021-01143-6
48. Pavey TG, Gomersall SR, Clark BK, Brown WJ. The validity of the GENEActiv wrist-worn accelerometer for measuring adult sedentary time in free living. *J Sci Med Sport*. 2016;**19**(5):395–399. https://doi.org/10.1016/j.jsams.2015.04.007
49. Hewitt L, Okely AD, Stanley RM, Batterham M, Cliff DP. Validity of GENEActiv accelerometer wear and nonwear time for use in infants. *J Phys Act Health*. 2021;**18**(5):488–494. https://doi.org/10.1123/jpah.2019-0486
50. Dal EV, Snaphaan L, Bouwmeester S, de Kort Y, Bongers I. Testing a single-case experimental design to study dynamic light exposure in people with dementia living at home. *Appl Sci*. 2021;**11**(21):10221. https://doi.org/10.3390/app112110221
51. Montoya FG, Peña-García A, Juaidi A, Manzano-Agugliaro F. Indoor lighting techniques: an overview of evolution and new trends for energy saving. *Energ Buildings*. 2017;**140**:50–60. https://doi.org/10.1016/j.enbuild.2017.01.028

Published by Oxford University Press on behalf of Sleep Research Society 2025. This work is written by (a) US Government employee(s) and is in the public domain in the US.